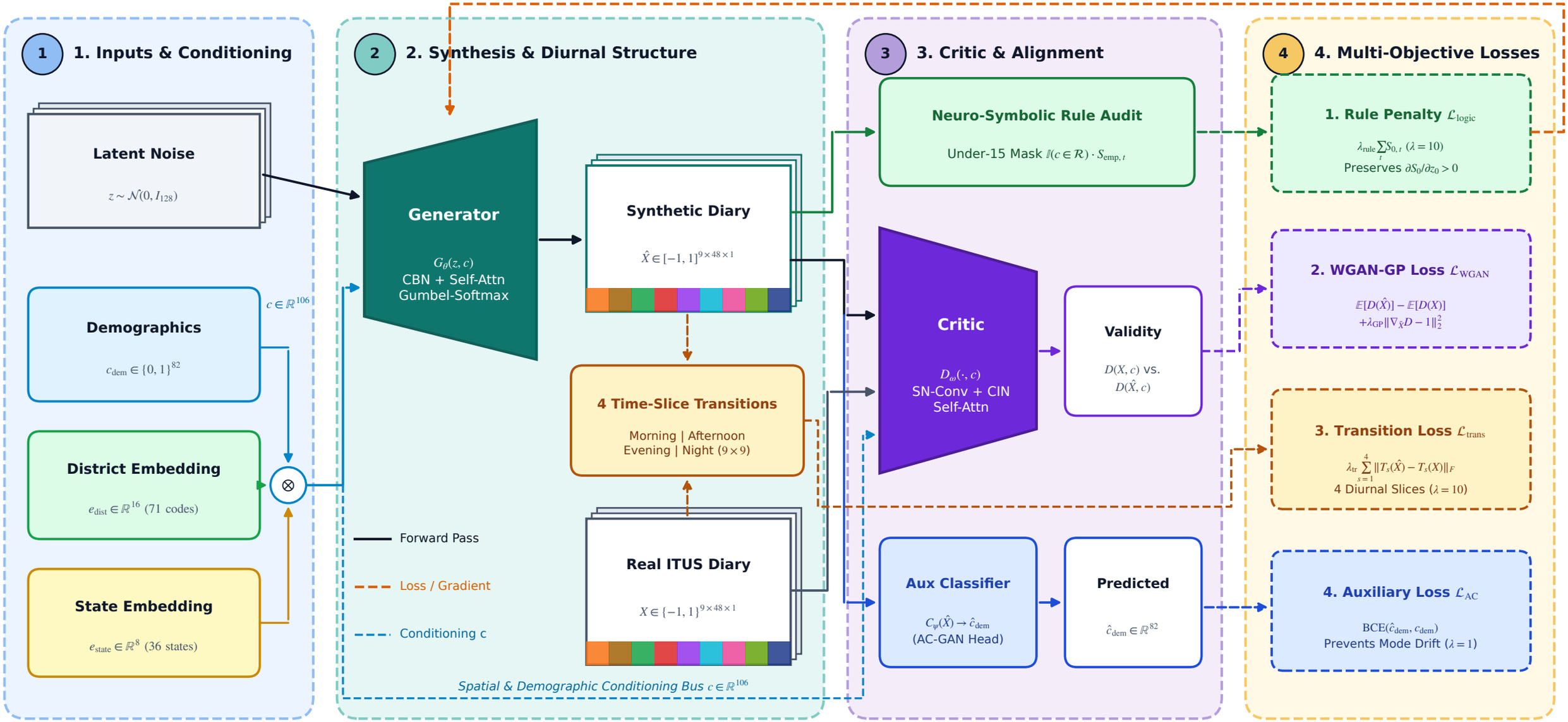

Backpropagated Multi-Objective Gradient Flow $\nabla_\theta(\mathcal{L}_G + \lambda_{tr}\mathcal{L}_{trans} + \lambda_{AC}\mathcal{L}_{AC} + \lambda_{rule}\mathcal{L}_{logic})$
1
1. Inputs & Conditioning
Latent Noise
$z \sim \mathcal{N}(0, I_{128})$
Demographics
$c_{dem} \in \{0,1\}^{82}$
$c \in \mathbb{R}^{106}$
District Embedding
$e_{dist} \in \mathbb{R}^{16}$ (71 codes)
State Embedding
$e_{state} \in \mathbb{R}^{8}$ (36 states)
2
2. Synthesis & Diurnal Structure
Generator
$G_\theta(z, c)$
CBN + Self-Attn
Gumbel-Softmax
Synthetic Diary
$\hat{X} \in [-1, 1]^{9\times48\times1}$
4 Time-Slice Transitions
Morning | Afternoon
Evening | Night (9×9)
Real ITUS Diary
$X \in \{-1, 1\}^{9\times48\times1}$
Forward Pass
Loss / Gradient
Conditioning c
Spatial & Demographic Conditioning Bus $c \in \mathbb{R}^{106}$
3
3. Critic & Alignment
Neuro-Symbolic Rule Audit
Under-15 Mask $\mathbb{I}(c \in \mathcal{R}) \cdot S_{emp,t}$
Critic
$D_\omega(\cdot, c)$
SN-Conv + CIN
Self-Attn
Validity
$D(X, c)$ vs.
$D(\hat{X}, c)$
Aux Classifier
$C_\psi(\hat{X}) \to \hat{c}_{dem}$
(AC-GAN Head)
Predicted
$\hat{c}_{dem} \in \mathbb{R}^{82}$
4
4. Multi-Objective Losses
1. Rule Penalty $\mathcal{L}_{logic}$
$\lambda_{rule} \sum_t S_{0,t}$ ($\lambda = 10$)
Preserves $\partial S_0 / \partial z_0 > 0$
2. WGAN-GP Loss $\mathcal{L}_{WGAN}$
$\mathbb{E}[D(\hat{X})] - \mathbb{E}[D(X)]$
$+\lambda_{GP}\|\nabla_{\hat{X}} D - 1\|_2^2$
3. Transition Loss $\mathcal{L}_{trans}$
$\lambda_{tr} \sum_{s=1}^{4} \|T_s(\hat{X}) - T_s(X)\|_F$
4 Diurnal Slices ($\lambda = 10$)
4. Auxiliary Loss $\mathcal{L}_{AC}$
BCE($\hat{c}_{dem}, c_{dem}$)
Prevents Mode Drift ($\lambda = 1$)